

Poverty & Inequality:

A Data-Driven Analysis Using World Bank Development Indicators

Data Analytics Capstone Project — AnalystLab Africa Internship

1. Objective / Problem Statement

Poverty and income inequality remain among the most pressing global development challenges. This project applies a complete data analytics workflow — extraction, cleaning, analysis, and visualization — to World Bank development data in order to answer three questions: How has global poverty changed over time? How unequally is income distributed across world regions? And what relationship, if any, exists between a country's economic output and its poverty rate? The goal is to translate raw indicator data into insights that can support evidence-based policy discussion.

2. Dataset Description

The dataset used is the World Bank's World Development Indicators (WDI), the flagship collection of global development statistics covering over 200 countries and territories across multiple decades. From the full WDI catalogue, six indicators relevant to poverty and inequality were selected for this analysis:

- Gini index — a standard measure of income inequality (0 = perfect equality, 100 = maximum inequality)
- Income share held by the lowest 20% of earners
- Income share held by the highest 20% of earners
- Poverty headcount ratio at \$3.00 a day (2021 PPP), % of population
- Unemployment, total (% of total labor force), modeled ILO estimate
- GDP per capita (current US\$)

The raw source file contained 396,970 rows spanning all WDI indicators, countries, and years in wide format (one column per year). Rows include both individual countries and World Bank-defined aggregates (e.g. "World", "Sub-Saharan Africa", income-group classifications), which were retained and explicitly tagged to support both country-level and regional analysis.

3. Data Cleaning Process

The dataset was cleaned and prepared using Python (pandas) through the following steps:

- Filtered the full dataset down to the six indicators above, using pattern matching to resolve naming inconsistencies in the raw WDI file — for example, the poverty line indicator had shifted from a \$2.15/day (2017 PPP) to a \$3.00/day (2021 PPP) threshold in this WDI release, and the unemployment

indicator required distinguishing the standardized "modeled ILO estimate" from the non-comparable "national estimate" variant.

- Reshaped the data from wide format (one column per year) to long format (one row per country–indicator–year), producing 104,940 rows across 6 columns.
- Removed rows with no reported value, since indicator coverage varies significantly by country and year (household survey-based indicators such as poverty and Gini are not collected annually in every country).
- Converted the Year column to a whole-number type and removed exact duplicate rows.
- Merged in official WDI country metadata to tag each row as "Country" or "Aggregate" and to attach Region and Income Group classifications, enabling regional comparison.

After cleaning, the dataset was reduced to 33,191 valid rows and exported as a CSV file for use in Power BI.

4. Analysis Methodology

Analysis was conducted in two stages. First, Python (pandas) was used for all data extraction, filtering, reshaping, and cleaning steps described above. Second, the cleaned dataset was loaded into Power BI Desktop to build an interactive dashboard, including:

- Four slicers (Indicator Name, Region, Year, and Type) allowing dynamic filtering
- Six KPI cards displaying the global average for each of the six indicators
- A line chart tracking the global poverty rate trend over time
- A bar chart comparing the average Gini index across world regions
- A bar chart comparing the income share gap between the lowest and highest 20% of earners, by region
- A matrix table comparing GDP per capita and poverty rate across individual countries

A custom color theme (deep slate blue, warm amber, and an institutional off-white background) was applied throughout for a professional, report-ready presentation.

5. Key Findings

Average Gini Index (global)	37.33
Average Poverty Rate (\$3.00/day, 2021 PPP)	14.48%
Average Unemployment Rate (modeled ILO)	7.78%
Average Income Share — Highest 20%	44.48%
Average Income Share — Lowest 20%	6.71%
Average GDP per Capita	\$8,920

- Globally, the top 20% of earners hold an average of 44.48% of income, while the bottom 20% hold just 6.71% — a gap of nearly 38 percentage points.
- The poverty rate trend shows a sharp rise in early tracked data, followed by a sustained decline in more recent decades — suggesting meaningful global progress against extreme poverty over time, despite regional unevenness.
- Latin America & Caribbean and Sub-Saharan Africa show the highest average Gini index of any region (approximately 48 and 44 respectively), making them the most unequal regions in the dataset.
- Europe & Central Asia shows the lowest average Gini index (approximately 33), indicating comparatively equal income distribution.
- The same two high-inequality regions — Latin America & Caribbean and Sub-Saharan Africa — also show the widest gap between top and bottom 20% income shares, reinforcing the Gini index findings.

6. Insights

- Inequality and income concentration are closely linked: regions with the highest Gini index also show the largest gap between what the richest and poorest 20% of the population earn, rather than these being two separate phenomena.
- Global poverty reduction has not been evenly distributed — while the overall trend is downward, the persistence of high Gini index scores in specific regions suggests that economic growth alone has not translated into equitable income distribution everywhere.
- Europe & Central Asia's comparatively low inequality suggests that stronger social protection systems and more progressive economic structures can coexist with sustained development, offering a possible reference point for higher-inequality regions.

7. Recommendations

- Policymakers in high-Gini regions, particularly Latin America & Caribbean and Sub-Saharan Africa, should prioritize progressive taxation and expanded social safety nets to narrow the income share gap between the richest and poorest households.
- Sustained investment in education and employment programs is recommended to continue supporting the declining global poverty trend observed in this analysis.
- Further country-level research is recommended to identify the specific policy and structural drivers behind regional Gini disparities, enabling more targeted interventions than region-wide averages alone can support.
- Dashboards of this kind should be maintained and refreshed as new WDI data is released, allowing stakeholders to track progress toward global goals such as UN Sustainable Development Goal 10 (Reduced Inequalities) over time.

8. Conclusion

This analysis of World Bank Development Indicators confirms that while global poverty has broadly declined over the tracked period, income inequality remains highly uneven across regions. Latin America & Caribbean and Sub-Saharan Africa stand out as the most unequal regions by both Gini index and income share gap, while Europe & Central Asia demonstrates that lower inequality is achievable alongside continued development. These findings point toward a clear policy priority: closing the gap between poverty reduction and equitable income distribution, particularly in the regions where the two indicators diverge most sharply.